

Hands-on-Lab: Generative AI Applications for Content Creators



Estimated time: 30 minutes

Welcome to the hands-on lab generative AI applications for content creators.

Learning objectives

After completing this lab, you will be able to:

- Demonstrate examples of creating content based on specific styles, tones, and target audiences through generative AI
- Demonstrate use cases of image generation through generative AI for content creators

Introduction

Content creators in different domains are leveraging generative AI to enhance efficiency, productivity, and creativity. Generative AI can enable improved content quality and increased consistency. Another important benefit of generative AI is enhanced personalization. Generative AI can enable the creation of messages, blogs, newsletters, and learning material tailored to specific audiences. In addition to text generation, content creators also leverage the image-generating capabilities of generative AI for diverse uses, such as creating designs and patterns and generating graphics, infographics, and logo designs.

In this lab, you'll work within the generative AI classroom to experiment with a few use cases for content creation and image generation.

About generative AI classroom lab

The generative AI classroom allows you to write and compare prompts to generate desired text with real-time chat responses. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

You will have the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen. You can interact with the language model using the message and chat fields.

The screenshot shows a web browser interface split into two main sections. The left section, titled "Instructions", contains the lab's title, estimated time (30 minutes), learning objectives, and an introduction. The right section, titled "Generative AI Classroom", features a "PROMPT INSTRUCTIONS" box with a placeholder text, a "How to use AI Classroom" section with a link to IBM watsonx.ai, and a chat area with a "Type your message" input field and a "Start chat" button. The interface is powered by watsonx.

Exercise 1: Creating personalized content

Generative AI has evolved to be more adaptive across various domains, catering to specific styles, tones, and target audiences. This empowers content creators across different fields to generate personalized and customized content tailored to the unique preferences and needs of their audience.

In this exercise, you will explore a few use cases to demonstrate creating personalized content in the domains of marketing and education.

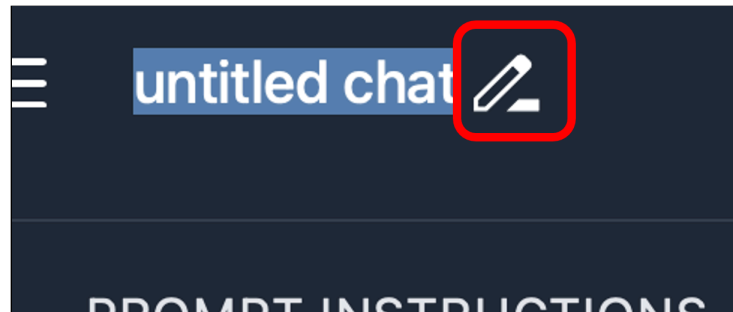
Note: For these use cases, you must ensure that the prompts provided are specific and clear and provide necessary instructions about the desired style, tone, and target audience.

Step 1: Set up the AI classroom

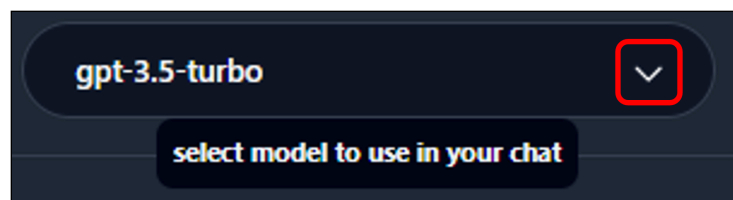
As a first step, you must set up your AI classroom for a better learning experience.

Before beginning the exercise, you must:

1. Name the chat: Use the pencil icon available on the top-left corner of the right pane to name the chat.

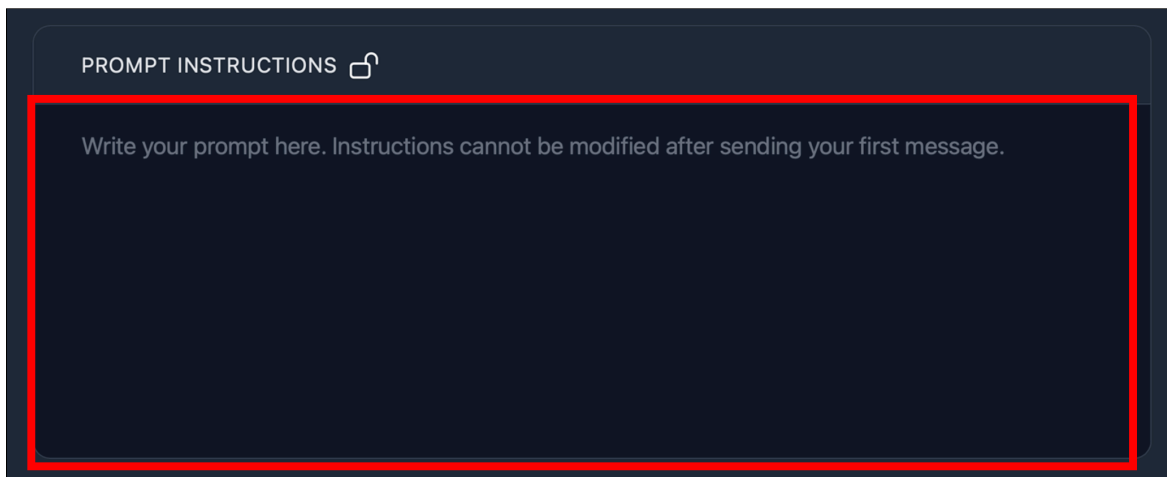


2. Choose the model: Use the dropdown option from the top-right corner of the right pane and select gpt-4o-mini for the text generation exercise.

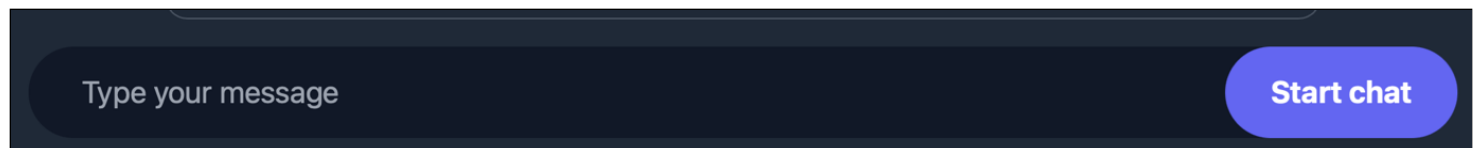


Note: gpt-3.5-turbo is the default model selected when you open your generative AI classroom.

3. Give prompt instructions: Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.



4. Type your message: Use the text box available at the bottom of the page to write the prompts and converse with the chat system.



Step 2: Generate the desired content

Example 1: Social media caption

Let's consider the scenario of a health and wellness brand. The marketing team wants to generate a social media caption to promote their protein-rich product for increased metabolism and high energy.

Target Audience: Weight-conscious women on social media

Style and tone: Positive, motivational, and casual

1. You must ensure that the prompt provided to the generative AI tool provides necessary instructions about the desired style, tone, and target audience. Let's enter the following prompt:

Generate an engaging social media caption highlighting our protein-rich product designed for women who prioritize weight control. Emphasize the benef

PROMPT INSTRUCTIONS 

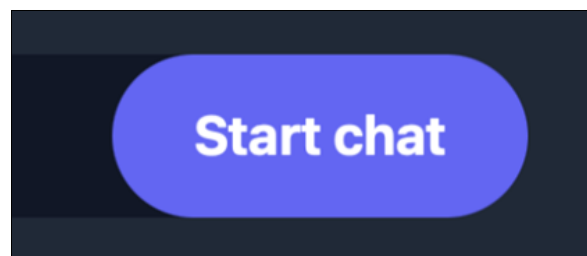
Write your prompt here. Instructions cannot be modified after sending your first message.

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Generate an engaging social media caption highlighting our protein-rich product designed for women who prioritize weight control. Emphasize the benefits of increased metabolism and sustained energy. The tone should be positive, motivational, and customized to inspire confidence in our female audience.

Start chat

2. Click Start chat to generate the output.



3. The received output is shown in the following screenshot. If you are not satisfied with the output, you can select Regenerate response to generate a new output.

Generate an engaging social media caption highlighting our protein-rich product designed for women who prioritize weight control. Emphasize the benefits of increased metabolism and sustained energy. The tone should be positive, motivational, and customized to inspire confidence in our female audience.

"Fuel your day with our protein-packed product specially formulated for women who are focused on weight control. Boost your metabolism and maintain sustained energy levels with every serving. Take charge of your health and confidence today! #ProteinPower #MetabolismBoost #SustainedEnergy #ConfidenceBoost"

 Regenerate response

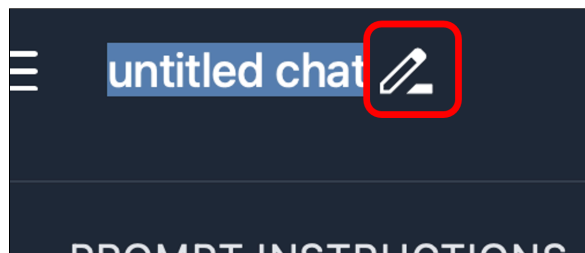
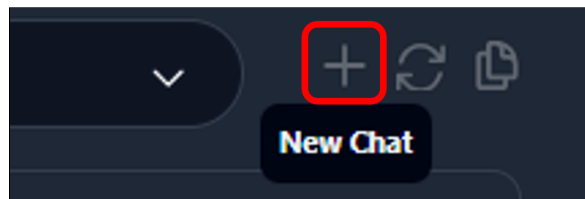
Example 2: Empathetic customer support mail

A customer support executive seeks support from generative AI to draft an email response to a customer who has raised a complaint about their order that has been delayed beyond the expected delivery date.

Style and tone: Empathetic, formal, and assuring

Target audience: A concerned and upset customer

1. Select New Chat in the top right corner of your generative AI classroom lab, and then select the pencil icon to name the chat.



2. While typing the prompt, you need to ensure that it includes details about the desired style and tone and what content you want to include in the mail; for example, an explanation for the delay and an assurance to the customer that their concerns are being addressed. In the new chat of your generative AI classroom, let's enter the following prompt:

Compose a polite and empathetic email response to a customer who has raised a complaint about their order being delayed beyond the expected delivery

PROMPT INSTRUCTIONS

Write your prompt here. Instructions cannot be modified after sending your first message.



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Compose a polite and empathetic email response to a customer who has raised a complaint about their order being delayed beyond the expected delivery date. Apologize for the inconvenience, provide a brief explanation for the delay, and assure the customer that their concerns are being addressed. Offer any relevant compensation or solutions and thank the customer for their patience.

Start chat

3. Click Start chat. Based on the prompt, the following output was received:

Subject: Apology for Delayed Order and Resolution

Dear [Customer's Name],

I hope this email finds you well. I am writing to sincerely apologize for the delay in delivering your order. I understand how frustrating it can be to not receive your purchase within the expected timeframe, and I want to assure you that we are taking immediate action to address this issue.

The delay was caused by the unexpectedly high demand for the product, which led to a backlog in our shipping department. Rest assured, we are working tirelessly to expedite the processing and shipment of your order.

To compensate for the inconvenience, we would like to offer you a [specific discount or free shipping on your next order]. Additionally, if you would like to cancel your order and receive a full refund, please let us know, and we will process it promptly.

We value your business and want to make things right for you. Thank you for your understanding and patience during this time. If you have any further questions or concerns, please do not hesitate to reach out to our customer service team at [contact information].

Thank you for choosing [Company Name]. We appreciate your support.

Warm regards,

[Your Name]
Customer Service Team
[Company Name]

Example 3: Financial advice newsletter for young professionals

A financial advisor at an investment firm is working on writing a financial advice newsletter, targeted at young professionals. The newsletter focuses on encouraging young professionals to save and invest wisely, empowering them to build a secure financial future.

Style and Tone: Practical, informative, and empowering

Target Audience: Young professionals navigating finances

1. In a new chat of your generative AI classroom lab, let's enter the following prompt:

Generate a practical and informative newsletter article targeted at young professionals, providing tips and advice on how to save and invest wisely t

PROMPT INSTRUCTIONS

Write your prompt here. Instructions cannot be modified after sending your first message.

 Want to elevate your AI learning experience? Check out [IBM watsonx.ai](https://ibm.com/watsonx.ai) now!

Generate a practical and informative newsletter article targeted at young professionals, providing tips and advice on how to save and invest wisely to build a secure financial future. The tone should be empowering, encouraging readers to take control of their finances. Include insights on budgeting, smart investment strategies, and practical steps for financial empowerment. Aim to inspire financial literacy among young professionals navigating their financial journey.

Start chat

2. Click Start chat. Based on the prompt, the following output was received:

Title: "Empower Your Financial Future: Tips for Young Professionals on Saving and Investing Wisely"

Dear Young Professionals,

As you embark on your journey towards financial independence, it's important to equip yourself with the knowledge and tools to make informed decisions about your money. Building a secure financial future starts with smart saving and investing strategies. Here are some practical tips and advice to help you take control of your finances and pave the way towards a brighter financial future.

1. Budgeting: The first step towards financial empowerment is creating a budget. Start by tracking your expenses and identifying areas where you can cut back. Set realistic savings goals and prioritize building an emergency fund to cover unexpected expenses. Budgeting allows you to take control of your spending and make informed decisions about where your money goes.

2. Smart Investment Strategies: Investing is a key component of building wealth over time. Consider diversifying your investment portfolio to spread risk and maximize returns. Explore options such as index funds, mutual funds, and exchange-traded funds (ETFs) that offer a balanced mix of assets. Remember to regularly review and adjust your investment strategy based on your financial goals and risk tolerance.

3. Practical Steps for Financial Empowerment: Take advantage of employer-sponsored retirement plans, such as 401(k) or Roth IRA, to save for retirement and benefit from tax advantages. Consider automating your savings and investments to ensure consistent contributions towards your financial goals. Educate yourself on financial literacy through books, online resources, and workshops to stay informed and make informed decisions about your money.

By taking control of your finances and making smart choices about saving and investing, you can build a secure financial future and achieve your long-term goals. Remember, financial empowerment is a journey that requires dedication and discipline, but the rewards are well worth the effort. Start today and pave the way towards a brighter financial future.

Empower yourself, take control of your finances, and secure your financial future. Your journey towards financial independence starts now.

Best regards,

[Your Name]

Example 4: Script for a YouTube video for elementary school students

Consider a YouTube channel that creates educational content for children. The development team wants to generate a script through generative AI for a YouTube video explaining the water cycle for elementary school students.

Style: Informative, interactive, and visually engaging

Tone: Enthusiastic, approachable, and age-appropriate

Target audience: Elementary school students (ages 5-10)

1. In a new chat of your generative AI classroom lab, let's enter the following prompt:

Create an engaging and educational script for a YouTube video aimed at elementary school students (ages 5–10) to explain the water cycle. The style s

PROMPT INSTRUCTIONS 

Write your prompt here. Instructions cannot be modified after sending your first message.

 Want to elevate your AI learning experience? Check out [IBM watsonx.ai](https://ibm.com/watsonx.ai) now!

Create an engaging and educational script for a YouTube video aimed at elementary school students (ages 5–10) to explain the water cycle. The style should be informative, visually engaging, and interactive, incorporating age-appropriate graphics and animations. The tone should be enthusiastic and approachable. The script should cover key concepts of the water cycle, such as evaporation, condensation, and precipitation, and include interactive elements to encourage participation and learning. Consider incorporating fun facts, simple experiments, and engaging visuals.

Start chat

2. Click Start chat. Based on the prompt, the following output was received:

Title: The Amazing Water Cycle Adventure!

(Opening shot of a cheerful animated character standing in front of a colorful backdrop)

Character: Hi everyone! Welcome to our exciting journey through the water cycle! Today, we're going to dive into the world of water and learn all about how it moves and changes form. Are you ready? Let's go!

(Scene transitions to a simple animation of the sun shining on a body of water)

Character: Our adventure begins with the sun's rays shining down on the water, causing it to heat up and turn into water vapor. This process is called evaporation. Can you say "evaporation"?

(Interactive text pops up on the screen encouraging viewers to repeat "evaporation")

Character: That's right! Evaporation is when water turns into invisible vapor and rises up into the sky.

(Transition to a cute cloud character forming in the sky)

Character: As the water vapor rises, it cools down and turns back into tiny water droplets, forming clouds. This is called condensation. Can you say "condensation"?

(Interactive text pops up on the screen encouraging viewers to repeat "condensation")

Character: Great job! Condensation is when water vapor turns back into liquid water, creating fluffy clouds in the sky.

(Transition to a raindrop character falling from a cloud)

Character: When the clouds get too heavy with water droplets, they release the water back down to Earth in the form of rain, snow, or hail. This is called precipitation. Can you say "precipitation"?

(Interactive text pops up on the screen encouraging viewers to repeat "precipitation")

Character: Awesome! Precipitation is when water falls from the clouds and replenishes our rivers, lakes, and oceans.

(Ending shot of the character waving goodbye)

Character: And that's the amazing water cycle! I hope you had fun learning about how water moves and changes form. Remember, the water cycle is always in motion, so keep an eye out for these processes in the world around you. Thanks for joining me on this adventure! See you next time!

(End screen with a link to subscribe and a fun water cycle quiz for viewers to test their knowledge)

Exercise 2: Image-generating applications of generative AI for content creators

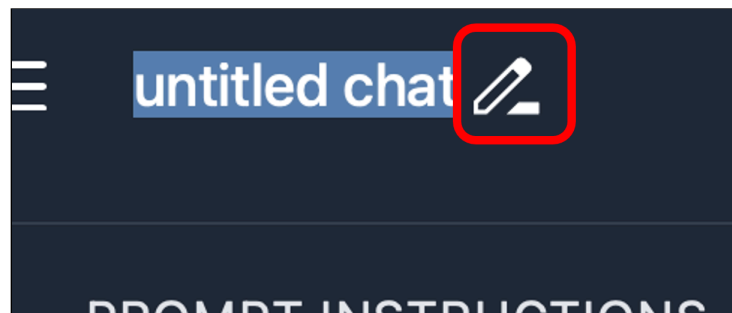
Content creators in different domains can use the image-generating capabilities of generative AI to enhance creativity and efficiency. Content creators in the digital art field can use generative AI to inspire and augment their artistic creations. Fashion designers leverage generative AI to experiment with new patterns, textures, and designs. Content creators in the marketing domain can use generative AI to create visuals for campaigns and posts.

For this exercise, let's consider a fictitious startup called EcoSol, which focuses on environmental awareness. EcoSol wants to run a social media campaign to raise awareness about plastic pollution and encourage their audience to adopt eco-friendly alternatives.

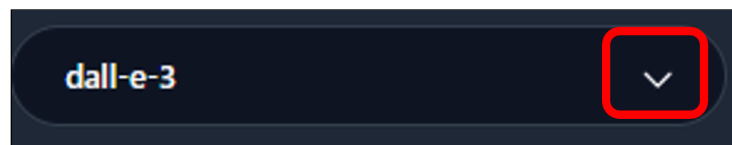
Step 1: Set up the generative AI classroom

Before beginning the exercise, you must:

1. Name the chat: Use the pencil icon available on the top-left corner of the right pane to name the chat.



2. Select the model: Both Dall-E 2 and Dall-E 3 models are used for generating images. For this exercise, let's use the dall-e-3 model. Use the dropdown option from the top-right corner of the right pane and select dall-e-3 for the image generation exercise.



3. Give prompt instructions: Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.

PROMPT INSTRUCTIONS

Write your prompt here. Instructions cannot be modified after sending your first message.

4. Type your message: Use the text box available at the bottom of the page to write the prompts and converse with the chat system.

Type your message

Start chat

Step 2: Generate a visually impactful image

Think of a context and draft a statement or prompt for the desired image you want from the chatbot. For better results, include specifics and details about the context of the required output.

1. Write your prompt or statement in the **Type your message** field at the bottom of the screen. Let's enter the following prompt:

Generate a visually impactful image that vividly illustrates the harmful effects of plastic pollution on oceans and marine life.

You can copy the prompt using the copy icon on the right side of the box.

Generate a visually impactful image that vividly illustrates the harmful effects of plastic pollution on oceans and marine life.

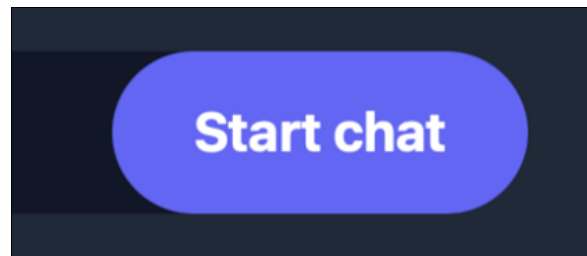
Start chat

Note: For better results, you can include additional information or specific instructions about the context of the required output in the Prompt Instructions field.

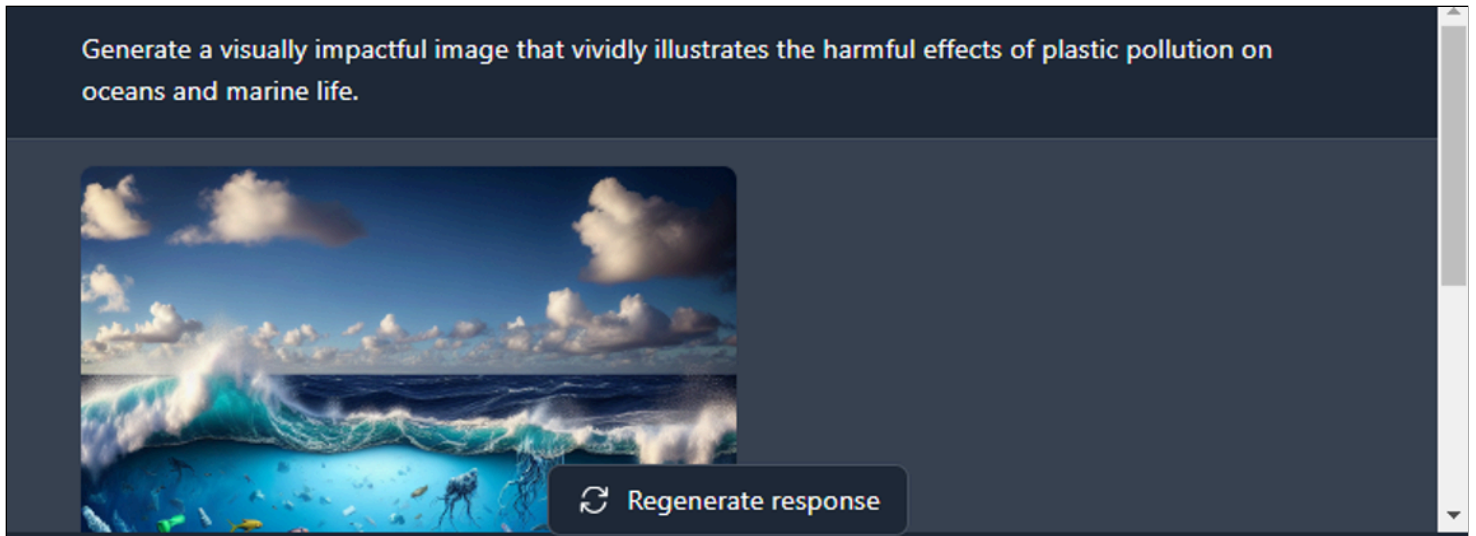
For example,

Include different types of fish.

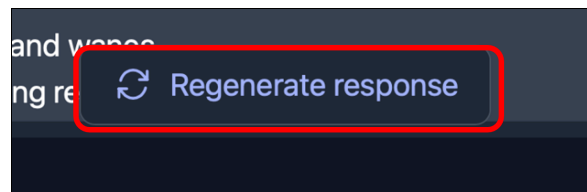
2. Now, click **Start chat** to send the message to the chatbot.



3. After a few seconds, the images will be generated. The image that was generated at our end as the output is displayed here.



4. If you are unsatisfied with the response, click **Regenerate response**, and the chat system will show a revised response for your original prompt.



5. You can save the generated image by right-clicking on it and then selecting **Save image as**. The actual image generated is shown below.



Step 3: Try yourself

Now, you can try to describe and generate images of your choice using the generative AI classroom.

Summary

Congratulations! You just completed the hands-on lab Generative AI Applications for Content Creators.

In this lab, you explored a few use cases of generative AI to create personalized content based on specific styles, tones, and target audiences through generative AI. You also worked with an image-generating tool to demonstrate use cases of image generation through generative AI for a marketing team.

Author(s)

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